



ARTS ET MÉTIERS INSTITUTE OF TECHNOLOGY,
LILLE CAMPUS

Preliminary Design of a Glider Wing

COURSE: PJT

GLIDER WING

Students:

POUILLY Léo
BREUVART Kylian

Instructor:

DAZIN Antoine

Contents

I	Introduction	2
1	Objectives of the session	2
II	Determination of the Incidence Angle	2
1	Method details	2
2	Expression of the lift coefficient	3
3	Calculation of the lift force	3
4	Incidence angle for the RAF 27 airfoil	3
5	Incidence angle for the CLARK Y airfoil	3
III	Comparison of the Two Models	5
1	Determination of the Lift-to-Drag Ratio	5
2	Determination of the Maximum Lift and Stall Angle	6
IV	Conclusion	7

I Introduction

1 Objectives of the session

The objective of this session is to perform the preliminary sizing of a glider wing and to calculate the incidence angle that meets the specifications listed below.

Specifications:

- $M = 500kg$: mass of the glider
- $V = 80km/h$: flight speed of the glider
- $L = 22m$: total wingspan of the glider
- $c = 1.25m$: chord length

We will therefore calculate this incidence angle for two different airfoil profiles and then determine which one best meets the specifications:

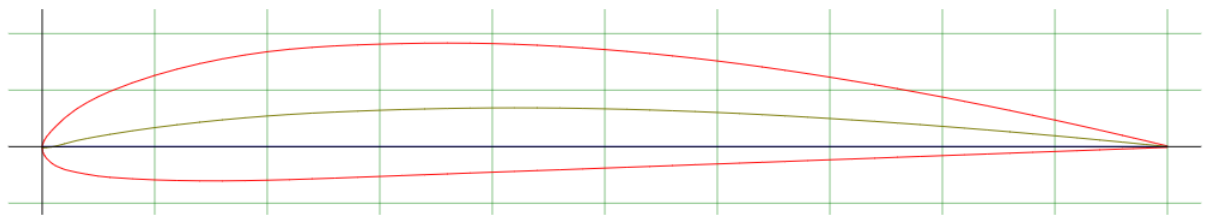


Figure I.1: CLARK Y Airfoil

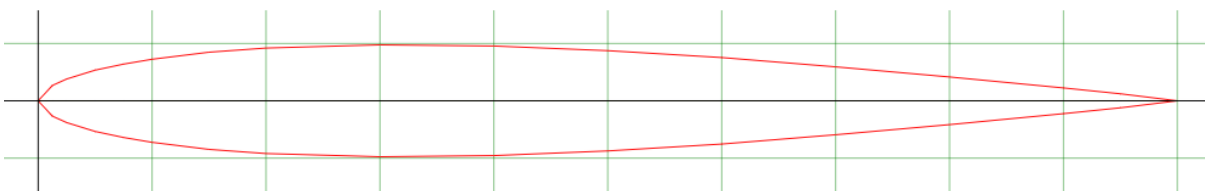


Figure I.2: RAF 27 Airfoil

II Determination of the Incidence Angle

1 Method details

The objective of this lab session is, for each airfoil, to determine the incidence angle that satisfies the specifications. To achieve this, we are provided with charts relating the incidence angle to the drag coefficient C_x and the lift coefficient C_z . Our first goal is therefore to determine these coefficients. To do so, we will first calculate the Reynolds number corresponding to our study and then determine the lift coefficient by applying the fundamental principle of dynamics to our glider wing.

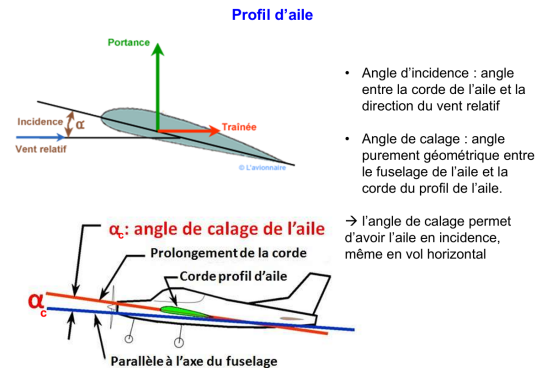


Figure II.1: Diagram of a wing profile

2 Expression of the lift coefficient

We assume that the glider is flying perfectly horizontally. Thus,

$$V_{relative\ wind} = V_{flight}$$

We can determine the Reynolds number:

$$Re = \frac{V \cdot c}{\nu} = 1.78 \times 10^6$$

with

- $V_{aircraft} = 80\ km/h = 22.2\ m/s$
- $\nu_{air} = 15.6 \times 10^{-6}\ m^2/s$
- $c = 1.25\ m$

It can be observed that for a Reynolds number on the order of 10^6 , the graphs showing C_z as a function of C_x are quite similar.

3 Calculation of the lift force

By applying the fundamental principle of dynamics along the z-axis, we find that:

$$F_z = M \cdot g$$

This expression represents the limiting case of the lift force, since for the glider to take off, we must have $F_z > Mg$. Using the expression of the lift force, we obtain the expression of the limiting lift coefficient:

$$C_z = \frac{2Mg}{\rho V^2 L c} = 0.63$$

4 Incidence angle for the RAF 27 airfoil

The drag coefficient can be determined using the graph above, where we read $C_x = 0.1$.

We obtain an incidence angle of: $\alpha = 5^\circ$

5 Incidence angle for the CLARK Y airfoil

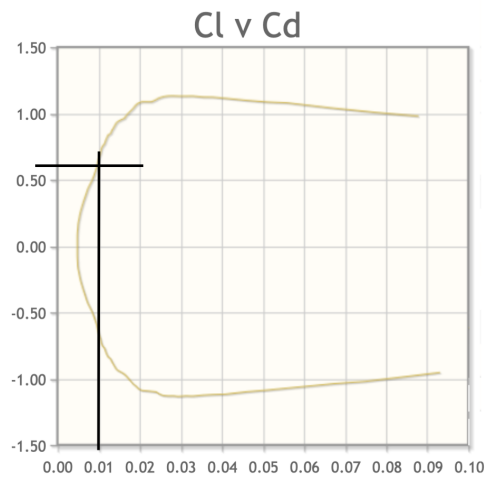


Figure II.2: Lift coefficient as a function of the drag coefficient

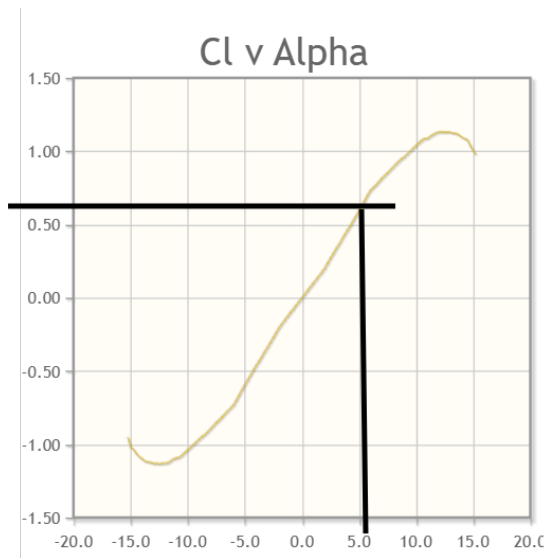


Figure II.3: Lift coefficient as a function of the incidence angle

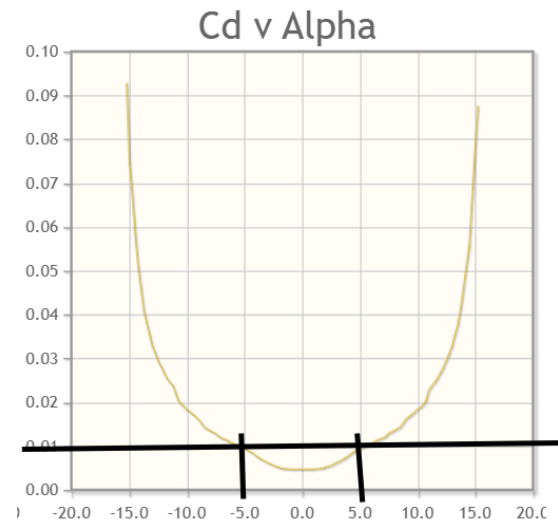


Figure II.4: Drag coefficient as a function of the incidence angle

Similarly, we find a drag coefficient $C_x = 0.1$, and from the graphs below, we deduce an incidence angle of: $\alpha = 2.5^\circ$

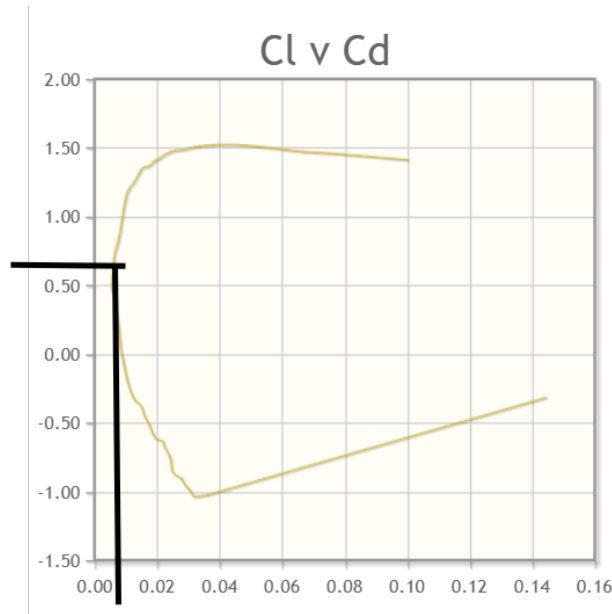


Figure II.5: Lift coefficient as a function of the drag coefficient

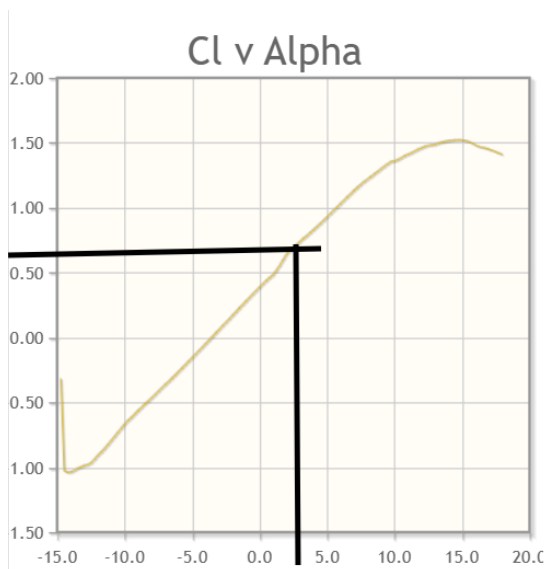


Figure II.6: Lift coefficient as a function of the incidence angle

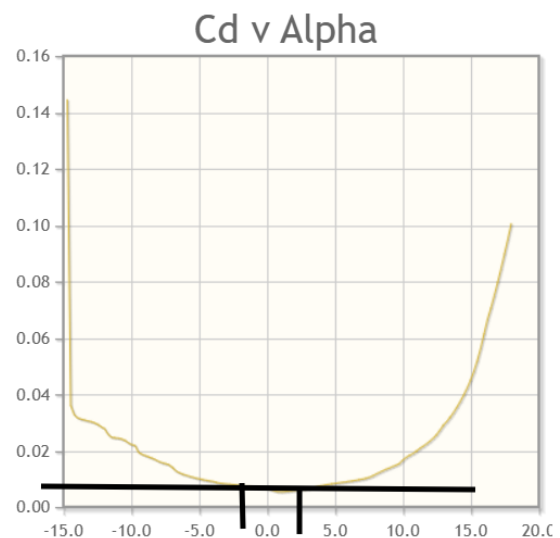


Figure II.7: Drag coefficient as a function of the incidence angle

III Comparison of the Two Models

1 Determination of the Lift-to-Drag Ratio

To compare the two wings, we will evaluate the lift-to-drag ratio, which expresses the relationship between the lift and drag forces. The maximum lift-to-drag ratio corresponds to the point of maximum efficiency (best distance-to-energy-consumption ratio), and therefore constitutes a key criterion in our choice of airfoil.

$$f = \frac{C_z}{C_x}$$

By assuming horizontal flight, which allows us to establish a correspondence between the angle of attack and the incidence angle, we can graphically determine the lift-to-drag ratio for each profile.

For the RAF 27 airfoil:

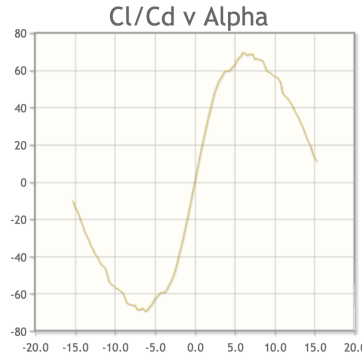


Figure III.1: Lift-to-drag ratio as a function of the incidence angle for the RAF 27 airfoil

We find $f = 64$ and a maximum lift-to-drag ratio $f_{max} = 70$.

For the CLARK Y airfoil:

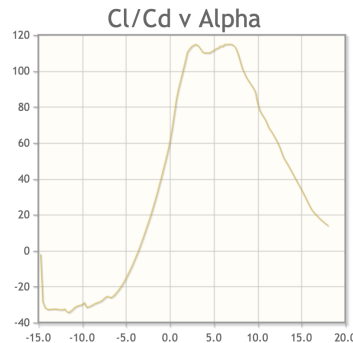


Figure III.2: Lift-to-drag ratio as a function of the incidence angle for the CLARK Y airfoil

We find $f = 112$ and a maximum lift-to-drag ratio $f_{max} = 115$.

2 Determination of the Maximum Lift and Stall Angle

The lift coefficients are determined graphically.

For the RAF 27 airfoil, from Figure II.3:

We find $C_{z,max} = 1.1$ for $\alpha_{stall} = 12.5^\circ$.

For the CLARK Y airfoil, from Figure II.6:

We find $C_{z,max} = 1.5$ for $\alpha_{stall} = 14.7^\circ$.

We can then determine the maximum lift force $F_{z,max}$ for each airfoil:

For the RAF 27 airfoil: $F_{z,max} = \frac{1}{2} \times C_{z,max} \times \rho \times V^2 \times L \times c = 8.95 \text{ kN}$

For the CLARK Y airfoil: $F_{z,max} = \frac{1}{2} \times C_{z,max} \times \rho \times V^2 \times L \times c = 12.2 \text{ kN}$

IV Conclusion

	Angle de calage	Finesse	Finesse maximale	Portance maximale	Angle de décrochage de l'aile	Angle de décrochage du planeur
RAF 27	5°	64	70	8,95kN	12,5°	7,5°
CLARK Y	2,5°	112	115	12,2kN	14,7°	12,2°

Figure IV.1: Summary table

The Clark Y airfoil has both a higher horizontal lift-to-drag ratio and a higher maximum lift-to-drag ratio than the RAF 27 airfoil. The maximum lift-to-drag ratio of the Clark Y is 64.29% greater than that of the RAF 27, while its horizontal lift-to-drag ratio is 75% higher. Its maximum lift coefficient as well as its stall angle are also superior to those of the RAF 27 profile.

However, machining the Clark Y airfoil can be more complex and therefore more expensive.

We therefore decide to choose the Clark Y airfoil.